

Introduction to Modeling for Formal Verification

Today's goals



Module 1

Introduction to Model Checking

Module 2

Modeling

Module 3

Modeling System Composition

Modeling Concurrency Using Transition Systems

✓

Concurrency and Interleaving

Video • 10 min

✓

Concurrency and Interleaving: Examples

Video • 8 min

✓

Principles of Model Checking - Section 2.2.1, (Page 35-39)

Reading • 30 min

Modeling Concurrency Using Transition Systems Quiz

Graded Assignment • 15 min

Modeling Communications Via Shared Variables

Interleaving Program Graphs

Video • 11 min

Critical Actions and Atomicity

Video • 10 min

Peterson's Mutual Exclusion Algorithm

Video • 12 min

Principles of Model Checking - Section 2.2.2, (Page 39-47)

Reading • 30 min

Modeling Communications Via Shared Variables Quiz

Graded Assignment • 20 min

Modeling Communications Via Handshaking

Synchronization and Handshaking

Video • 12 min

Example: Railroad Crossing

Video • 13 min

Principles of Model Checking - Section 2.2.3, (Page 48-52)

Reading • 30 min

Modeling Communications Via Handshaking Quiz

Graded Assignment • 10 min

Synchronous Parallelism

Synchronous Product - Part 1

Video • 10 min

Synchronous Product - Part 2

Video • 10 min

The State Space Explosion Problem

Video • 9 min

Principles of Model Checking - Section 2.2.6, (Page 73-75)

Reading • 20 min

Synchronous Parallelism Quiz

Graded Assignment • 20 min

Principles of Model Checking - Section 2.2.1, (Page 35-39)



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Textbook - Principles of Model Checking - Chapter 2.2.1 (page 35-39)

Section 2.2 explains a number of communication models for modeling parallel systems. Section 2.2.1 considers systems that are composed of independent processes, and explains the construction of a system transition system by interleaving the transition systems for consisting processes without communications.

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✓ Completed

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